

Beyond a Good Story

When and for whom does persuasive loan language actually speed up funding?

Team Cultural Blend
University of Technology Sydney
Kiva loan data · 1.45 million real loans · 2016–2025

The questions

1

When a **loan's story** leans on **family, competence, or urgency** does it get funded **faster**?

2

OR Does the result shift by WHO the borrower is and WHERE & WHEN they posted?

Why it matters: Loan story framing is the one lever a platform can coach. Loan size, sector and geography are fixed at listing

How the 1.45M loan investigation is structured

ACT I · CONTEXT

SLIDES 4–5

The Operating Reality

Why narrative advice is pitched, and what really happens on the ground.

- The Platform Dilemma: Text is the only unsealed lever at listing; platforms want to coach it.
- Market Slowdown: Same-day funding collapsed from 46% to 30% post-pandemic.
- Capital Reality: 96.4% pre-disbursed; partner working capital waits, not borrowers.

ACT II · INFERENCE

SLIDES 6–9

Empirical Stress-Test

Testing narrative claims against 1.45M loans with econometric rigor.

- Methodological Screen: 64 of 120 coefficients collapse under country-clustering.
- Structural Dominance: Amount, terms, and sector beat storytelling by ~10×.
- Narrative Scrutiny: Urgency dissolves ($p \approx 0.49$); family rests on 2-country screen.
- Topic Modeling: Topics track loan purpose (water vs farming; 9× swing).

ACT III · STRATEGY

SLIDES 10–12

The Strategic Playbook

High-ROI platform decisions, borrower realism, and epistemic bounds.

- 4-Quadrant Playbook: Kill generic tips, A/B test family framing, fix structural lag.
- Borrower Realism: Set cash-flow expectations; do NOT game gender or story.
- Scientific Bounds: Observational associations, completed-loans survival truncation.

APPENDIX · AUDITS

SLIDES 14–17

Methodological Audits

Full numerical proofs and statistical architecture for defense.

- Two-Country Problem: The conservative $t(1) = 12.7$ heuristic screen in full.
- Predictive vs Causal: Why SHAP importance cannot certify policy rules.
- The ~10× Comparison: Decile monotonicities and correlation breakdown.
- 24h Early Warning: LightGBM classifier architecture (ROC AUC 0.8997).

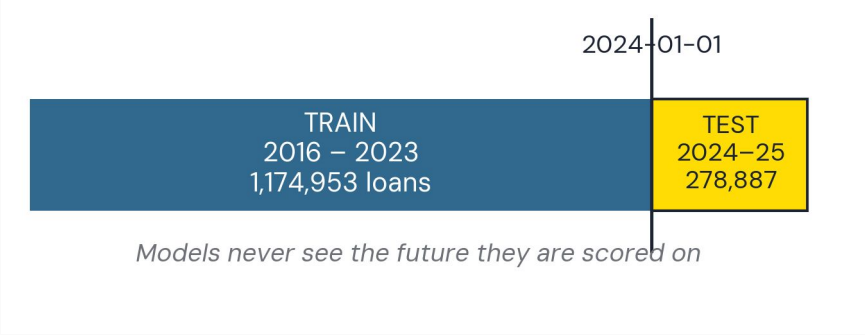
PORTFOLIO METHODOLOGY & REPRODUCIBILITY STACK · 1,453,846 REAL LOANS (2016–2025)

Econometric OLS Duration Modeling · Binomial GLM · Country-Clustered Standard Errors · Conservative Few-Cluster $t(1)=12.7$ Screen · NMF Topic Extraction (8 Components) · LightGBM Predictive Benchmark (Strict Chronological Split, Zero Future Leakage)

How we stress-tested our findings

- Predictive claims: trained only on the past, tested only on loans posted 2024–2025. No peeking at the future.
- Framing claims: Every "significant" framing result went through two more tests, standard errors clustered by country, then a stricter few-cluster screen where it rested on a handful of countries.
- Across both explanatory models, 64 of 120 significant coefficients did not survive the country-clustered re-test. That is the point of the method.
- SHAP importance shown as complementary predictive evidence only. It cannot confirm or refute a statistical finding.

Figure 1. Chronological train/test split.



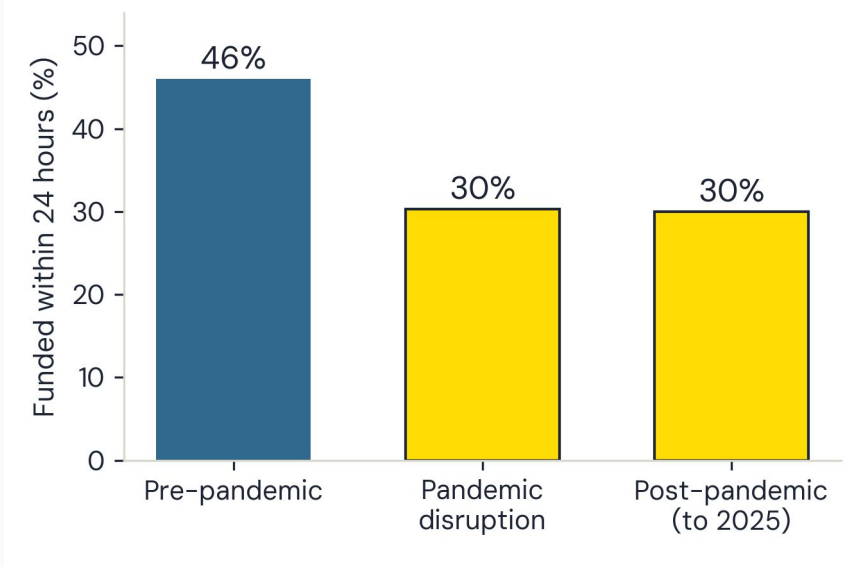
Notes: The 278,887 test loans stayed sealed during training - the models never see the future they are scored on.

A slower market is the operating reality — and it is partner capital that waits.

46.0% → 30.3% → 30.0% funded within 24 hours

- Pre-pandemic, almost half of all loans funded within a day.
- Since 2020: under a third — and four years on, still not recovered. Partner capital now waits longer, not the borrower: 96% of loans are already disbursed before the page goes live.
- 589,823 loans before vs 565,474 after — not a thin sample, and the slowdown persists to date.

Figure 2. Share of loans funded within 24 hours, by analysis period.



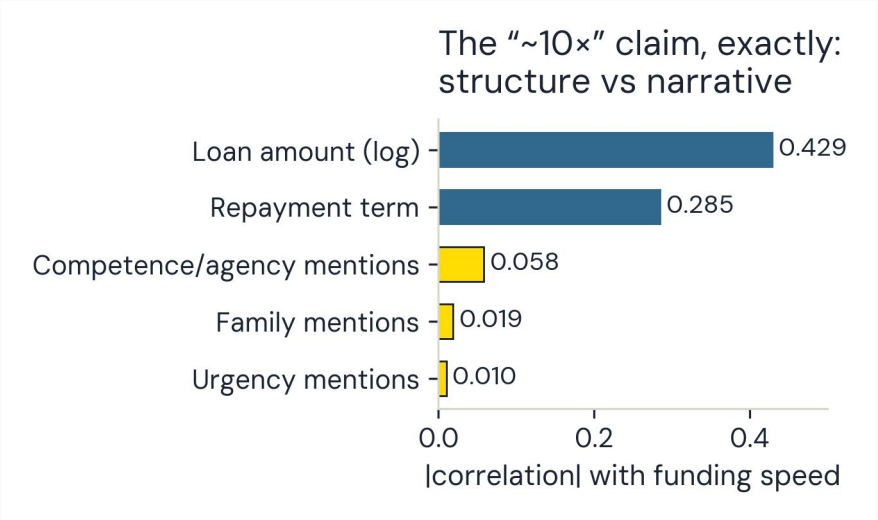
Notes: The drop is 16 points; the 'recovery' is -0.3. Exact shares 46.0% / 30.3% / 30.0% (589,823 / 298,549 / 565,474 loans).

Structural loan factors beat storytelling by an order of magnitude

- Loans posted by women fund in a median 2.3 days; by men, 7.7
- Loan amount + repayment terms: linked to funding speed ~10× more strongly than any single narrative choice. Bigger ask, longer on the platform.
- Sector alone spans well over an order of magnitude in speed; country gaps are wider still — 0.2 to 20.5 days.
- These spreads are associations, not causes.

Figure 3. Correlation with funding speed: structural vs narrative features.

The "~10x" claim, exactly: structure vs narrative



Notes: Loan amount correlates ~7x stronger with speed than the best narrative signal (0.429 vs 0.058).

Storytelling coaching: no robust evidence it moves the needle.

No narrative result

is robust enough to support a writing recommendation

- Urgency: looked universal (p<0.001) — collapses under country clustering (p≈0.49). Not recommended.
- Family: corrected test → Middle East (Palestine+Yemen) & Central America (Honduras+Nicaragua) same direction in every fit — but 2 countries each; few-cluster t(1) screen bar is 12.7, not 1.96: p≈0.06–0.21. A hypothesis, not a finding.
- Sentiment: positive tone ↔ slower funding, but significance flips between specifications — genuinely open.
- SHAP: no family / agency / urgency feature in the model's top 15 ⇒ complementary evidence, not confirmation.

Figure 4. Average same region family-framing slope (duration model).

Within-region family-framing slope (duration model): none pass the screen

region (ctys)	loans	slope	cl. p	few-cl. p	verdict
Africa (27)	610,368	-0.010	0.554	t(26) 0.559	n.s.
Asia (12)	738,191	+0.034	0.054	t(11) 0.080	n.s.
C. America (2)	59,391	-0.062	<0.001	t(1) 0.065	norm-ref only
Middle East (2)	14,946	-0.124	<0.001	t(1) 0.075	norm-ref only
N. America (1)	7,559	+0.011	0.062	not estimable	1 country
Oceania (4)	23,385	+0.016	0.631	t(3) 0.663	n.s.

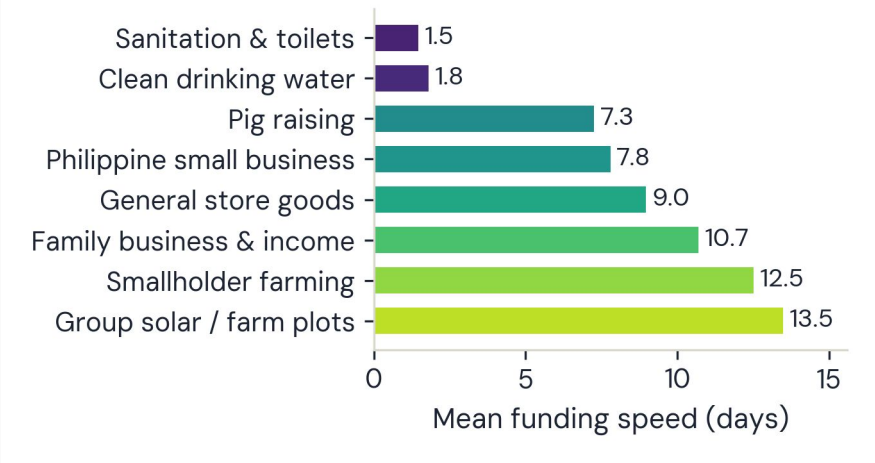
Notes: For a two-country region the significance bar is t(1) = 12.7, not 1.96 - nothing clears it.

The themes come down to loan purpose, not persuasion.

From 1.5 days
To 13.5 days
across story topics — a ninefold swing

- **Topic modeling** on the descriptions (TF-IDF + NMF, 8 topics, not keyword counts) surfaces coherent themes: sanitation, clean water, pig raising, family business, smallholder farming.
- The largest single gap anywhere in the analysis, but a topic mostly encodes **what the loan is FOR**. Structure again, not persuasion.

Figure 5. Mean funding speed by story theme.



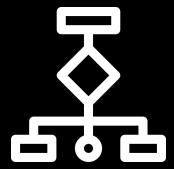
Notes: Water and sanitation stories fund in under 2 days; group farming loans take ~2 weeks to fund - themes track what the loan is FOR.

What this means in practice for the Kiva team



DON'T ship writing tips — a platform-wide 'add urgency' nudge would be built on a result that doesn't survive testing.

DO run a stratified A/B test of family framing by country: Palestine, Yemen, Honduras and Nicaragua. Randomised writing prompt at listing, outcome time to fund, analysed country by country.



DO review the structural gaps (programme / sector / region / gender) — they are ~10x the size of any wording effect.



PROTOTYPE, then re-validate: the 24h classifier ranks well among eventually-funded loans (AUC ≈ 0.90, strictly future data), but expired/withdrawn listings aren't in the data, so an early warning flag first needs all posted listings, a defined target, and retraining on that population.

What this means in practice



What to expect when listing

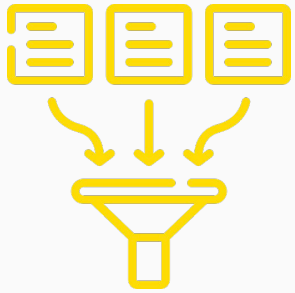
For borrowers: what to expect — not what to game

Profile	Female-posted loans fund in a median 2.3 days vs 7.7 male-posted — a gap for Kiva to investigate, not a lever for borrowers
Loan amount & repayment terms	The two strongest speed predictors — but partner-set against a real need. Plan timing around them; don't shrink the ask
Sector	Speed varies ~13× by sector (0.9–12.1 days) — set expectations by category; don't switch categories to chase speed

What this can't tell us



Association, never causation. No borrower was randomly assigned a writing style.



Measures how fast a **FUNDED** loan funds, not whether a loan gets funded at all.



Framing measured with transparent, simple rules, not every nuance of persuasive writing.

“In this data, the story barely registers.
The structure carries the signal.”

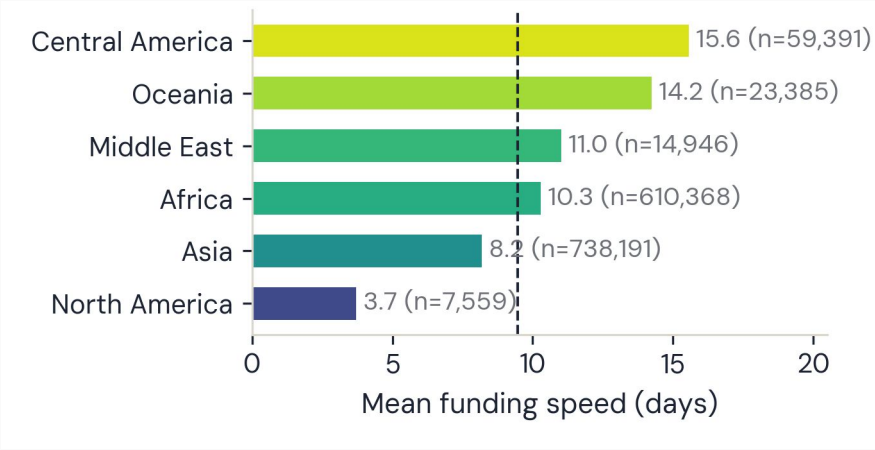
Thank you —
questions



The two-country problem, in full

- Pooled categories: Middle East = Palestine + Yemen (14,946 loans); Central America = Honduras + Nicaragua (59,391); North America = Haiti alone.
- Few-cluster screen p (duration / 24h / notebook fit): ME 0.12 / 0.21 / 0.08 · CA 0.06 / 0.14 / 0.07 — same direction in every fit, none significant; t(1) critical value 12.7.
- North America: single country — between-country uncertainty not estimable at all.
- The screen is a conservative heuristic: it can downgrade a claim, never certify one. Next step: country-stratified A/B test in those four markets.

Figure 6. Mean funding speed by region.



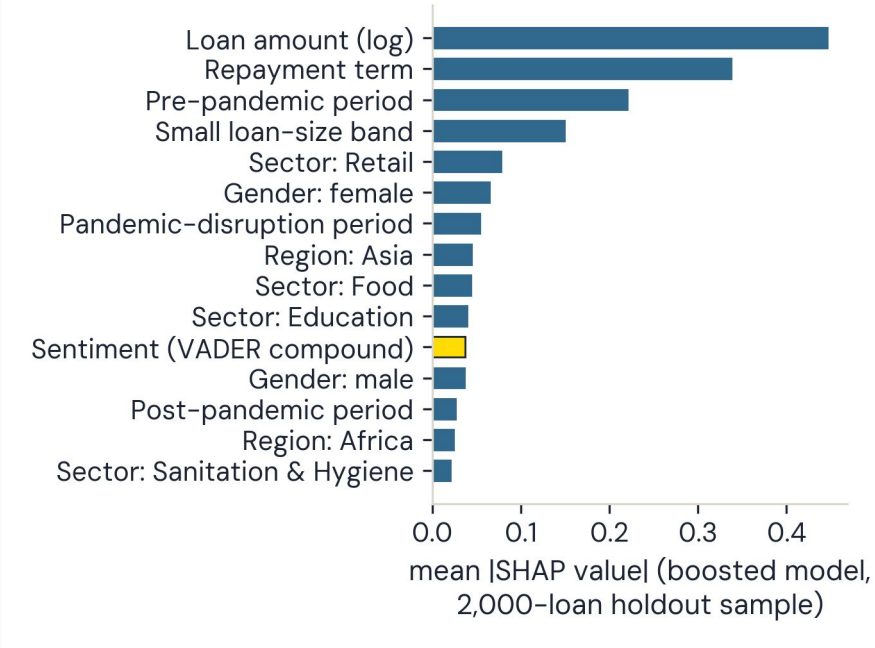
Notes: North America is Haiti alone (n = 7,559) - fastest region, and exactly why single-cluster inference is off the table.

Predictive weight ≠ statistical robustness

- SHAP importance from the boosted forecasting model: structure fills the top ranks (amount, term, period, size band, sector, gender).
- No family, agency, or urgency feature reaches the top 15; sentiment is 11th despite its disputed significance.
- Complementary predictive evidence only — a different model, no region interactions; it cannot confirm or refute any coefficient's sign or uncertainty.

Figure 7. Top 15 features by mean |SHAP| value, boosted model.

SHAP top 15: sentiment (11th) is the only narrative feature



Notes: Ten structural features outrank the first narrative one.